Replication and Extension of Card & Krueger (1994)

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1. Introduction

This project replicates and extends Card and Krueger (1994), "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania," published in the *American Economic Review* 84(4), 772-793.

The paper asks a simple but economically important question: does raising the minimum wage reduce employment? On April 1, 1992, New Jersey raised its state minimum wage from $4.25 to $5.05 per hour, an 18.8% increase and, at the time, the highest state minimum wage in the United States. Neighboring eastern Pennsylvania kept its minimum wage at $4.25. This setting provides a natural experiment: restaurants in New Jersey form the treatment group, and restaurants in eastern Pennsylvania form a credible comparison group, since the two regions share similar labor markets, seasonal patterns, and product markets.

Card and Krueger surveyed 410 fast-food restaurants (Burger King, KFC, Roy Rogers, and Wendy's) in both states before the increase (February-March 1992) and again about eight months after (November-December 1992). Comparing employment changes across the two states, they found no evidence that the minimum-wage increase reduced employment. In fact, full-time-equivalent (FTE) employment rose slightly in New Jersey relative to Pennsylvania. This result challenged the standard competitive model's prediction that a binding minimum wage lowers employment, and it became one of the most influential findings in labor economics.

The goal of this project is twofold. First, I replicate the paper's core empirical results, namely Tables 2, 3, and 4 and Figure 1, using the publicly available survey data. Second, I extend the analysis by examining heterogeneity in the employment effect: whether the effect differs depending on how binding the minimum wage was for a given store, on the chain, and on whether the store is company-owned or franchised.

The replication closely reproduces the paper's main results. The difference-in-differences (DID) estimate of the employment change is +2.75 FTE employees (paper: +2.76); the New Jersey dummy in the regression is +2.36 (paper: +2.33); and the wage-gap coefficient is +15.9 (paper: +15.65). The extension shows that the positive employment effect is concentrated in stores where the minimum wage was most binding and in franchise-owned stores.

The remainder of the report is organized as follows. Section 2 describes the data and sample construction. Section 3 presents the empirical design. Section 4 reports the replication results and compares them with the original paper. Section 5 presents the extension. Section 6 discusses validation and limitations, and Section 7 concludes.

2. Data and Sample Construction

Data source. The data are the public-use version of the Card and Krueger (1994) fast-food survey, distributed for this course (Package 01) as a single file, `public.csv`. The dataset contains 410 restaurants surveyed in two waves: wave 1 in February-March 1992 (shortly before the minimum-wage increase) and wave 2 in November-December 1992 (about eight months after). Each restaurant appears as one row, with wave-1 variables in plain

columns (e.g., `EMPFT`, `EMPPT`, `NMGRS`, `WAGE_ST`) and wave-2 variables in columns suffixed with `2` (e.g., `EMPFT2`, `EMPPT2`, `NMGRS2`, `WAGE_ST2`).

Sample composition. Of the 410 restaurants, 331 are in New Jersey (the treatment group) and 79 are in eastern Pennsylvania (the comparison group). Table 2 shows the distribution of stores by chain and ownership. Burger King is the most common chain (41.1% in NJ, 44.3% in PA), followed by Roy Rogers, KFC, and Wendy's. About one third of stores are company-owned (34.1% in NJ, 35.4% in PA).

Variable construction. Following the paper (Table 2 note), full-time-equivalent (FTE) employment is defined as:

> FTE = full-time workers + managers and assistant managers + 0.5 x part-time workers

I construct FTE employment separately for each wave:

- `emptot1 = EMPFT + NMGRS + 0.5 x EMPPT` (wave 1, before the increase)
- `emptot2 = EMPFT2 + NMGRS2 + 0.5 x EMPPT2` (wave 2, after the increase)

The dependent variable in the employment regressions is the change in FTE employment, `emptotd = emptot2 - emptot1`.

Handling store closures. The second wave did not re-interview every store in the first wave. Following the paper's treatment (Table 3 note), I handle non-response and closures as follows, based on the second-wave status variable `STATUS2`:

`STATUS2`	**Meaning**	**Treatment of wave-2 FTE**
1	Answered	use reported values
3	Permanently closed	set to **0**
0, 2, 4, 5	Refused / renovation / highway construction / fire	set to **missing**

Setting wave-2 employment to zero for permanently closed stores ensures that the analysis measures the effect of the minimum wage on overall employment, not only on surviving establishments, an important feature of the paper's design.

Starting-wage groups and the wage gap. To compare stores by how binding the new minimum wage was, I classify New Jersey stores by their wave-1 starting wage: the low-wage group (exactly \$4.25 per hour, most affected, since the wage had to rise to \$5.05, an 18.8% increase), the midrange group (\$4.26-\$4.99 per hour), and the high-wage group (\$5.00 or more per hour, largely unaffected). I also construct the wage-gap variable used in the regression models:

- `gap = (5.05 - WAGE_ST) / WAGE_ST` for New Jersey stores initially paying below \$5.05
- `gap = 0` for Pennsylvania stores and for New Jersey stores already paying at least \$5.05

The wage gap measures the proportional wage increase a store must grant to comply with the new minimum. It equals zero for stores that were not constrained by the policy.

Analysis sample. For the regression models (Table 4), the analysis is restricted to stores with non-missing FTE employment in both waves and a valid wave-1 starting wage. This yields 365 stores, close to the paper's 357. The small difference reflects undocumented sample-screening choices in the original paper (see Section 6).

The mean and standard deviation of the dependent variable (change in FTE employment) are -0.252 and 8.749, respectively, close to the paper's -0.237 and 8.825.

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3. Empirical Design

The empirical strategy follows Card and Krueger (1994). Because the minimum-wage increase applied only to New Jersey, the natural approach is a difference-in-differences (DID) design that compares the change in employment in New Jersey with the change in Pennsylvania over the same period. Any common shock affecting both states, such as the regional recession of 1992, is differenced out, so the DID estimate isolates the effect of the minimum-wage increase.

3.1 Difference-in-Differences

The basic DID estimator is:

> DiD = (FTE_NJ,after - FTE_NJ,before) - (FTE_PA,after - FTE_PA,before)

where FTE denotes average full-time-equivalent employment per store. This compares the employment change at New Jersey stores (the treatment group) with the employment change at Pennsylvania stores (the comparison group). Under the assumption that, absent the policy, employment in the two states would have moved in parallel (the parallel-trends assumption), the DID estimate identifies the causal effect of the minimum-wage increase.

A useful specification check exploits variation within New Jersey. Stores that were already paying $5.00 or more per hour before the increase were largely unaffected by the new law, so they provide an alternative comparison group that is not subject to cross-state differences. If the DID estimate using Pennsylvania as the control group is similar to the estimate using high-wage New Jersey stores as the control group, the validity of the Pennsylvania comparison is supported.

3.2 Regression Models

The DID estimates are also expressed in regression form, which allows for control variables and alternative measures of treatment intensity. The baseline specification is:

> ΔE_i = a + c x NJ_i + ε_i (1a)

where ΔE_i is the change in FTE employment at store i between the two waves, and NJ_i is a dummy equal to 1 for stores in New Jersey. The coefficient c is the DID estimate, directly comparable to the simple difference in means.

An alternative specification replaces the New Jersey dummy with a continuous measure of how binding the minimum wage was at each store, the wage gap:

> ΔE_i = a' + c' x GAP_i + ε_i' (1b)

where GAP_i is the proportional increase in the starting wage required to reach the new minimum rate:

- `GAP_i = 0` for Pennsylvania stores
- `GAP_i = 0` for New Jersey stores with `WAGE_ST` >= $5.05
- `GAP_i = (5.05 - WAGE_ST_i) / WAGE_ST_i` for other New Jersey stores

Because GAP_i varies within New Jersey, the coefficient c' measures the employment response to the intensity of the minimum-wage increase, not just the cross-state difference. Card and Krueger show that GAP_i is a strong predictor of the actual wage change between the two waves (R-squared = 0.75), and that conditional on GAP_i there is no systematic difference in wage behavior between New Jersey and Pennsylvania stores.

The models are estimated by ordinary least squares. I report specifications with and without controls for chain type (three chain dummies) and company ownership, and, in the most demanding specification, controls for region (three regions of New Jersey and two regions of eastern Pennsylvania). The region controls absorb local demand shocks and identify the effect of the minimum wage from within-region variation between higher- and lower-wage stores.

3.3 Standard Errors

Standard errors are conventional OLS standard errors, following the original paper. All models include an unrestricted constant, which is not reported in the tables.

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4. Replication Results

This section reports the replication of Tables 2, 3, and 4 and Figure 1, and compares the results with the original paper.

4.1 Table 2: Descriptive Statistics

Table 2 reports means of key variables by state, separately for the two waves, with t-statistics for cross-state differences.

The store-type distribution matches the paper exactly: Burger King accounts for 41.1% of New Jersey stores and 44.3% of Pennsylvania stores; the corresponding figures are 20.5% and 15.2% for KFC, 24.8% and 21.5% for Roy Rogers, and 13.6% and 19.0% for Wendy's. Company-owned stores make up 34.1% of the New Jersey sample and 35.4% of the Pennsylvania sample.

In wave 1, average FTE employment was 20.44 in New Jersey and 23.33 in Pennsylvania, matching the paper (20.4 and 23.3). Starting wages were almost identical across states ($4.61 vs. $4.63). The average price of a full meal was higher in New Jersey ($3.35 vs. $3.04), while hours of operation and the prevalence of recruiting bonuses were similar.

By wave 2, average FTE employment was 21.03 in New Jersey and 21.17 in Pennsylvania. The average starting wage in New Jersey rose to $5.08, and 85.5% of New Jersey stores reported a starting wage of exactly $5.05, compared with only 1.3% in Pennsylvania. These patterns match the paper closely.

One variable differs from the paper: the percentage of full-time employees. My values are about 4 percentage points lower than the published figures (e.g., 28.6% vs. 32.8% in New Jersey in wave 1). This likely reflects a different denominator or weighting in the original paper, which is not fully documented. This does not affect the employment or wage results.

4.2 Table 3: Average Employment and Difference-in-Differences

Table 3 is the core of the paper. It reports average FTE employment before and after the increase, by state and by New Jersey starting-wage group.

Panel A (by state) reproduces the paper closely:

	PA	NJ	NJ - PA
FTE before	23.33	20.44	-2.89
FTE after	21.17	21.03	-0.14
Change	-2.17	0.59	**2.75**
Change (balanced)	-2.28	0.47	**2.75**
Change (temp. closed = 0)	-2.28	0.23	2.51

The DID estimate is **+2.75** FTE employees, essentially identical to the paper's +2.76. Employment fell in Pennsylvania by 2.17 FTE employees but rose in New Jersey by 0.59, so the relative gain in New Jersey is large and positive.

Panel B (within New Jersey) shows that the employment gain is concentrated in the stores that were most affected by the new minimum wage:

Group	FTE change (balanced)
Low wage ($4.25)	+1.21
Midrange ($4.26-$4.99)	+0.71
High wage (>=$5.00)	-2.16

The difference between low-wage and high-wage stores is **+3.36**, matching the paper exactly. This is a central result: stores that had to raise wages the most experienced the largest employment gains, while stores unaffected by the policy lost employment, just like stores in Pennsylvania.

4.3 Table 4: Regression Models

Table 4 reports reduced-form regressions of the change in FTE employment on either the New Jersey dummy or the wage-gap variable, with and without controls.

Model	Variable	Coefficient (SE)	Paper
(i)	NJ dummy	2.36 (1.15)	2.33 (1.19)
(ii)	NJ dummy + controls	2.33 (1.15)	2.30 (1.20)
(iii)	Wage gap	15.89 (5.95)	15.65 (6.08)
(iv)	Wage gap + controls	15.19 (6.08)	14.92 (6.21)
(v)	Wage gap + controls + region	12.03 (7.32)	11.91 (7.39)

The New Jersey dummy is +2.36 (paper: +2.33), implying that New Jersey stores gained about 2.4 more FTE employees than Pennsylvania stores. The wage-gap coefficient is +15.89 (paper: +15.65), implying that a store facing a 10% required wage increase gained about 1.6 additional FTE employees relative to an unaffected store. All coefficients are positive, as in the paper, and none indicates a negative employment effect. Adding

controls leaves the estimates almost unchanged, consistent with the paper's finding that observable store characteristics explain little of the variation.

The analysis sample contains 365 stores, compared with the paper's 357; the mean and standard deviation of the dependent variable (-0.252 and 8.749) are close to the paper's (-0.237 and 8.825).

4.4 Figure 1: Distribution of Starting Wages

Figure 1 plots the distribution of starting wages in New Jersey and Pennsylvania in February 1992 and November 1992. In February, the two distributions are very similar, with both states concentrated at $4.25. By November, the picture is dramatically different: nearly all New Jersey stores report a starting wage of $5.05, while the Pennsylvania distribution is essentially unchanged.

This figure provides direct evidence that the minimum-wage increase was binding: it pulled the entire lower tail of the New Jersey wage distribution up to the new minimum, without affecting Pennsylvania. It also shows no evidence of "spillovers" to higher-wage New Jersey stores.

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5. Extension: Heterogeneity in the Employment Effect

The baseline result shows a positive average employment effect of the minimum-wage increase. A natural question is whether this effect is uniform across stores, or whether it is concentrated in particular types of stores. I examine three dimensions of heterogeneity: how binding the minimum wage was, the chain, and ownership structure. All extensions use the same 365-store sample as Table 4.

5.1 Binding Intensity

I split New Jersey stores into three groups by their wave-1 starting wage:

Group	N	Mean FTE change	SE	t vs. high-wage
Low wage ($4.25)	94	+1.21	0.82	2.58 (p = 0.011)
Midrange ($4.26-$4.99)	132	+0.71	0.69	2.35 (p = 0.020)
High wage (>=$5.00)	67	-2.16	1.01	—

The employment gain is monotonically larger the more binding the minimum wage was. Stores initially paying exactly $4.25 gained 1.21 FTE employees on average, significantly more than unaffected stores paying at least $5.00 (which lost 2.16 FTE employees); the difference is significant at the 1% level. This confirms that the paper's aggregate result is driven by the stores the policy was intended to affect.

5.2 Chain Heterogeneity

I estimate the wage-gap regression separately for each chain:

Chain	N	Wage-gap coefficient	SE	p
Burger King	150	+21.58	10.50	0.042
Roy Rogers	91	+25.02	10.59	0.020

Chain	N	Wage-gap coefficient	SE	p
Wendy's	48	+15.91	21.85	0.470
KFC	76	-12.96	9.55	0.179

The positive wage-gap effect appears in the hamburger chains (Burger King and Roy Rogers), where it is statistically significant, and is absent in KFC, where the point estimate is actually negative (though not significant). This is consistent with the paper's observation that KFC stores are smaller, open fewer hours, and may absorb higher labor costs through other margins rather than through employment. The Wendy's estimate is positive but imprecise because of the small sample (N = 48).

5.3 Ownership Structure

Finally, I split the sample by ownership:

Type	N	Wage-gap coefficient	SE	p
Franchise	236	+22.17	8.16	0.007
Company-owned	129	+0.45	8.75	0.959

This is the strongest heterogeneity result. The positive employment effect is driven entirely by franchise-owned stores, where the wage-gap coefficient is +22.17 and highly significant (p = 0.007). Company-owned stores show essentially no effect (+0.45, p = 0.96). A plausible interpretation is that franchisees, as independent owner-operators whose profits are directly affected, adjust their operations more flexibly in response to a wage increase, whereas company-owned stores follow centrally determined staffing policies.

5.4 Summary

The extension shows that the positive employment effect of the minimum-wage increase is not uniform. It is concentrated in stores where the minimum wage was most binding, in hamburger chains, and, most strikingly, in franchise-owned stores. These findings complement Card and Krueger's aggregate result by identifying where the employment response comes from.

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6. Validation and Limitations

Validation. The replication reproduces the paper's main results closely. The DID estimate (+2.75 vs. +2.76), the low-versus-high-wage difference (+3.36 in both), the New Jersey dummy (+2.36 vs. +2.33), and the wage-gap coefficient (+15.89 vs. +15.65) are all within a few percent of the published values. The distribution of starting wages in Figure 1 matches the paper's key visual pattern. The robustness of the conclusion is supported by the within-New-Jersey comparison, which uses high-wage New Jersey stores as an alternative control group and yields a similar estimate.

Limitations. Three limitations should be noted.

First, the analysis sample contains 365 stores, compared with the paper's 357. The difference (about 2%) reflects undocumented sample-screening choices in the original paper. Because the sample construction is not fully specified in the paper, I could not reproduce the exact sample. The coefficients are nonetheless very close to the published values.

Second, the "percentage of full-time employees" in Table 2 differs from the paper by about 4 percentage points, likely because of a different denominator or weighting in the original definition. This variable does not enter the employment regressions, so it does not affect the main results.

Third, the extension relies on the same identification assumption as the replication (parallel trends). The within-New-Jersey comparison provides a specification check, but I cannot fully rule out state-specific demand shocks. The paper addresses this concern with region controls and an analysis of price changes; my extension does not repeat those tests.

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7. Conclusion

This project replicates and extends Card and Krueger's (1994) influential study of the employment effects of a minimum-wage increase. Using the publicly available survey data, I reproduce the paper's main results: employment in New Jersey rose relative to Pennsylvania following the 1992 minimum-wage increase, with a DID estimate of +2.75 FTE employees, essentially identical to the published +2.76. The regression models likewise reproduce the paper's findings, with the New Jersey dummy and the wage-gap variable both positive and close to the published values. The distribution of starting wages confirms that the policy was binding and that New Jersey stores adjusted their wages to the new minimum.

The extension adds a new dimension by showing that the positive employment effect is not uniform. It is concentrated in stores where the minimum wage was most binding, in hamburger chains, and especially in franchise-owned stores. This suggests that the employment response to a minimum-wage increase depends on firm structure, a dimension that the original paper did not emphasize.

Taken together, the replication and extension confirm Card and Krueger's central conclusion, namely that a moderate minimum-wage increase need not reduce employment, and add evidence on where the employment gains come from.

References

Card, David, and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): 772-793.